



OPEN Smart defense based on explainable stacked machine learning architecture for securing internet of health things with K-means clustering

Tanzila Saba¹, Amjad Rehman¹, Muhammad Mujahid¹, Amal Al-Rasheed², Shaha Al-Otaibi² & Abdallah Yousif³✉

The Internet of Health Things (IoHT) transformed current healthcare by facilitating real-time patient monitoring and remote diagnosis via networked medical equipment. The advanced prevalence of interconnected medical devices creates substantial vulnerabilities, rendering IoHT settings more prone to advanced cyberattacks due to their miniature dimensions and diverse nature. To address these challenges, this study presents Smart Defence-IoHT, a novel stacked architecture enhanced with KMeans clustering to effectively detect and classify multi-class cyber threats within IoT-based healthcare networks using the latest ECU-IoHT dataset. Firstly, key features related to cyber networks, such as packet time, length, and encoded protocol, source, and destination information, are extracted and normalized. KMeans clustering is applied to uncover latent structural patterns in the data by grouping similar instances, and the resulting cluster labels are appended as an additional feature to enrich the input space. The proposed stack model captures different aspects of the data distribution, and the output of classifiers is integrated through a meta model, which improves the predictions from the collective learning of stacking. Experimental results demonstrate that the proposed model performed outstanding, achieved 99.41% accuracy, 99.39% precision, 99.40% f1 score, 99.93% AUC, 98.72% MCC, and 0.0221 log loss for five different attacks such as ARP Spoofing, DoS, Nmap Port Scans, Smurf Attacks, and normal traffic. Precision, recall, and F1-score were included alongside accuracy to better capture the model's ability to detect attacks while minimizing false alarms, providing a more meaningful evaluation for real-world IoHT systems. The experimental results clearly demonstrate that the proposed method achieves a superior detection rate compared to existing approaches. Furthermore, we applied Explainable Artificial Intelligence (XAI) techniques including SHAP and LIME for highlighting the importance of each feature and provides interpretability and model trust that is important in healthcare.

Keywords IoHT, DoS attack, Clustering, Cybersecurity, XAI, Machine learning, Stacking, Smurf attack

Healthcare is changing rapidly in a digital environment. Digital technology is boosting healthcare and changing the way people and doctors manage their health¹. The IoT is a major healthcare advancement, and networks of medical devices, wearables, sensors, and software systems collect, measure, and analyse health data in real time². These devices monitor heart rate, blood pressure, blood sugar, and blood oxygen levels in and outside hospitals. IoT improves healthcare efficiency and personalisation³. Real-time data sharing will help diagnose, prevent, and treat chronic diseases. Wearable cardiac monitors can identify irregular heart rhythms and prevent heart attacks and strokes⁴. With the use of remote monitoring technologies, medical professionals can care for elderly or frail patients without physically visiting a clinic or hospital. This massive data set can help healthcare professionals improve their clinical decision-making, hospital readmission rates, resource utilisation, and healthcare

¹Artificial Intelligence and Data Analytics (AIDA) Lab, CCIS, Prince Sultan University, Riyadh, Saudi Arabia.

²Department of Information Systems, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, P.O. Box 84428, 11671 Riyadh, Saudi Arabia. ³Department of Computer Science, Sudan Technological University, Omdurman, Khartoum, Sudan. ✉email: abdallah.y@sudren.edu.sd

expenditures. IoT data can also inform clinical advancements, public health monitoring, and evidence-based treatment regimens when paired with big data analytics and AI⁵.

In healthcare, IoT technologies improve efficiency by supporting real-time patient monitoring and automating repetitive monitoring tasks. IoT has many benefits, but increasing adoption of new technology in healthcare creates cybersecurity and privacy risks⁶. Cybercriminals target IoT devices because they use open, scattered networks. Traditional IT infrastructure is small and complex, while IoT settings have many small, low-power devices, processors, and peripherals⁷. These devices can be widely distributed, but they have little or no built-in protection, making them easy to reach for unauthorised access, data theft, or physical inactivity. Hackers have exploited malfunctioning insulin or heart pumps, endangering lives. Sometimes criminals break into electronic health records to steal patient data, such as social security numbers, insurance information, and medical records, to sell on the dark web or use for fraud, scams, or extortion⁸. Extreme incidents have interrupted hospital networks, lost digital access, and prompted healthcare workers to return to work in person, frequently at tremendous risk to patients and staff. Such attacks raise ethical and legal issues regarding patient trust, organisational reputations, and accountability in medical technology management, as well as financial damages. Due to the amount and nature of health data, the healthcare business is one of the most vulnerable to cyberattacks⁹. Infrastructure and architecture are major IoT network security issues. With so many devices connected to private and public networks, it might be difficult to provide a comprehensive security solution. Furthermore, many of these products lack long-term software development capabilities¹⁰. Artificial intelligence (AI) and machine learning (ML) are used to protect IoT networks as a result of these difficulties¹¹. To spot attack trends, AI and ML models can analyse a variety of network traffic and device behavior data in real time. These models can identify irregularities that can point to data corruption, service theft, or security flaws. Because ML is scalable, it has a lot of potential. Before they inflict harm, such discoveries can uncover new vulnerabilities. The authors used DRNN and ML models for efficient and effective IDS for the IoMT. This technology can detect and predict unexpected cyber threats¹². It is challenging to forecast intelligent solutions in IoT setting.

AI is increasingly important in the IoHT, where interconnected medical devices collect and share sensitive patient data. It detecting cyber threats that could compromise patient safety. Within Healthcare 5.0, AI-driven IoHT fosters autonomous healthcare ecosystems that analyze complex interactions between health conditions, ensuring timely and personalized care¹³. To address significant cybersecurity weaknesses in IoHT, the paper¹⁴ suggests using a hybrid intrusion detection system that combines ANN with BLSTM and GRU designs. The Electronic Control Unit ECU-IoHT dataset, tailored to address the intricate security needs of IoHT, was used to conduct comprehensive model evaluations. Protecting data and networks must be a top priority when it comes to cybersecurity. In order to accomplish this, IoHT networks utilize IDS to keep an eye on data transfers, spot unusual behavior, and lessen the blow of cyberattacks. Kumar et al.¹⁵ said that time-series classification methods should be used to find possible attacks on IoHT networks. At first, they used neighborhood component analysis (NCA) with changes to the regularization setting to find the most important input features. They suggested two LSTM-based models that have the traits that were chosen. Real IoHT data were used to test these time-series classification models.

Problem statement

The integration of IoHT devices into modern healthcare systems is improving patient monitoring, personalized treatment, and overall healthcare efficiency. However, this rapid growth creates significant security vulnerabilities, as many IoT devices lack essential requirements such as encryption, authentication, and secure communication. Attacks on these systems can compromise not only sensitive medical data but also life-saving devices. Despite extensive research on these infrastructures, most devices still rely on DL or hybrid models, which require complex computing resources and often lack feature selection or preprocessing mechanisms, making them inherently unsuitable for IoT environments. Furthermore, the lack of publicly available healthcare-specific platforms further hinders their development, as existing platforms typically rely on IoT systems to address these vulnerabilities. The largest drawback is that a large number of IoT devices lack the processing capacity and battery life necessary to enable effective ML algorithms. Large, labeled datasets representing a range of common and uncommon behaviors are required to train trustworthy and generalizable ML models. This area is difficult to handle in the healthcare industry because of security concerns, device types, and usage patterns. Our study presents the ECUI-oHT dataset specifically designed to represent real-world clinical trials and help develop anomaly detection methods.

Contributions

This study employed ECU-IoHT dataset for the detection of various attacks using stacked model. Because, IDS datasets are disruptive, they are rarely integrated into complex interaction patterns, and healthcare environments. Communication security in existing IoHT devices is undoubtedly lacking, especially in light of the numerous vulnerabilities and data intrusions. The ECU-IoHT dataset addresses this challenge by incorporating realistic threat scenarios designed for the IoT and running them on a secure testbed. Unlike traditional network data infrastructure, ECU-IoHT highlights security risks specific to medical devices and connected healthcare systems, such as identity DoS, data theft, and device interruption. This is essential for evaluating medical malpractice prevention strategies when both patient safety and data integrity are at risk. By focusing on IoT disruption, the dataset provides methods that accurately reflects the challenges of managing digital healthcare systems. The study makes the following contributions:

- This study proposed a smart defense-IoHT-based, explainable stackable machine learning architecture for the detection of cybersecurity attacks on the IoHT. The model captures diverse attack patterns and improves detection performance by combining multiple models with a meta-estimator.

- A unique aspect of this work is the use of KMeans clustering to uncover latent behavioral patterns in network traffic data. The ECU-IoHT dataset is preprocessed and applies K-means clustering that is useful for dealing with complex data network monitoring, enriching the dataset and providing the model with unsupervised insights that enhance classification performance.
- The proposed framework is trained and evaluated against existing approaches, demonstrating its effectiveness through a comparative analysis. We conducted a thorough performance analysis using multiple evaluation metrics such as precision, precision, recall, F1 score, AUC, MCC, log loss, and Kappa.
- We applied SHAP and LIME techniques to enhance model transparency and trustworthiness, which are especially critical in healthcare. These explainers helped interpret the decisions made by the stacked model, enabling the identification of key features that influenced specific predictions. The proposed framework achieves better detection accuracy by analyzing an enormous amount of data.

The rest of the paper is structured as follows: “[Literature](#)” section explores recent related work for attack detection in IoHT, “[Materials and Methods](#)” section develops and describes the proposed methodology, “[Results and Discussion](#)” section presents the results and discussion, and “[Conclusion and Future Directions](#)” section concludes the study.

Literature

The study¹⁶ suggested images and the MobileNet design of CNNs to come up with a feature representation method that can better find cyberattacks in IoHT. The study used network traffic records from the ECU-IoHT dataset, the suggested method might find complicated patterns that point to intrusion. MobileNet-CNN's fast computation makes it possible to launch IoHT in real time even when resources were limited. Their approach was better than the current gold standard because it lowers the number of false positives and negatives and achieves a 99.1% classification accuracy, a 99.00% precision, a 97.1% recall, and a 98.4% F1-score. The results show that image-based deep learning models have a lot of potential as a way to find and get rid of threats before they happen. An XNN architecture¹⁷, author suggested to find middlebox risks in hospital IoT. XNN worked better than the baseline models. Adding real-time machines and reinforcement learning to this investigation could make the system work better and help the healthcare industry in the long run.

The study¹⁸ presented DNN-based method for protecting IoT-enabled medical devices by making it easier to find and stop intrusions in IoHT devices. The suggested method would use multi-class classification instead of binary class classification, which doesn't work against DoS attacks, ARP spoofing, Nmap attacks, and Smurf attacks. The ECU-IoHT dataset, which is related to healthcare, is another way that the work is judged. Based on the tests, the suggested DNN-based method has a much higher true detection rate and a much lower false detection rate than the present system. The authors utilized Federated Learning (FL) and Long Short-Term Memory (LSTM) in their research on predictive modeling for attack detection, which achieved satisfactory performance. The suggested model combines FL's decentralised design with advanced feature selection algorithms such as RUSBoost to make cyberthreat detection more accurate and protect sensitive healthcare data¹⁹. To deal with the big security risks that IoHT brings, this study offers a mixed system for finding intrusions that used DL. ANN with a Bidirectional Long Short-Term Memory (BLSTM) and a Gated Recurrent Unit (GRU) design would be used for the platform. The model, which was changed to meet the strict security requirements of IoHT, was tested fully using the ECU-IoHT dataset. The system does a great job in multiclass settings, with a perfect F1 score for recall, accuracy, and precision in binary classifications. These findings show that advanced AI techniques may help reduce false positives and improve detection accuracy²⁰.

DL model for smart healthcare data identification and cyberthreat detection was developed in study²¹. To find the important parts of IoMT, it first shows a combined ensemble learning method that gets rid of extraneous features and lowers the chance of the classifier becoming too good at its task. Scaled inputs were used by 1D-CLSTM, which have been suggested for classifying cyberthreats. The idea of random undersampling boosting has been used to solve problems like imbalance classification. The authors test the suggested model against cutting-edge techniques such as 1D-CNN, LSTM, and GRU, and used a lot of different performance measures to see how well it works. As shown in the paper²², a CNN-BiLSTM mixed deep learning model detect and classify intrusions in the Internet of Health using the most recent ECU-IoHT dataset. During all tests, the suggested solution had an amazing accuracy level, consistently doing better than a number of well-known techniques in the field of IoHT malware detection.

The objective of the collection was to help people who work in healthcare security figure out how attacks work and build strong defences. The ECU-IoHT dataset was used in this study to use advanced methods like RNN and CNN to detect and predict weaknesses in the healthcare system. AI-driven plans to help the shift of companies that focus on fixing problems while keeping private medical records safe. CNN achieved 98.9% accuracy for parallel classification and RNN achieved 98.9% accuracy for multi-class classification. These results show that the models did a great job²³. IoHT created a huge amount of time-related data and system log information, which makes it very hard to find threats quickly. The authors first, to pick the most important input features, they used neighborhood component analysis (NCA) with a changed regularisation value. These are the two LSTM-based models, they suggest based on the traits that were picked: Two ways are used to find cyberattacks: directed acyclic graph-based (DAG-LSTM) and projected layer-based (PL-LSTM). With real-life IoHT data, they compared the suggested models (like DAG-LSTM and PL-LSTM) to other time-series classification models (like LSTM, Bi-LSTM), the GRU, and others. With an average accuracy of 92.04% in testing and a training accuracy of 99.89%, their tests show that the DAG-LSTM was the most accurate¹⁵. Another research showed a federated, decentralized, and scalable deep learning framework that can find different types of attacks on healthcare data. In a decentralized client-server federated learning method, ECU-IoHT data was used to teach a deep learning model called GRU. Different kinds of hacking can be found by this model²⁴.

The suggested method streamlines the ideas of space and time by limiting the features to just ten out of forty-six. Using the CIC-IoT-2023 dataset, the authors achieved a multi-classification accuracy of 98.94% using the suggested technique²⁵. A gray-wolf optimizer (GWO) was used to pick the most attack-relevant features, and a soft voting mechanism was used to integrate the predictions from four fundamental classifiers: RF, SVM, DT, and XGBoost. Results were obtained by employing CNN. Using the CIC-IoT-2023 dataset, this model achieves an outstanding accuracy of 99.15%²⁶. The summary of literature review is shown in Table 1.

Materials and methods

This section explains the proposed method, which uses the ECU-IoHT dataset with 111,207 examples of both normal network traffic and attacks in the healthcare field. It is a resource for the healthcare security community to analyse attack behaviours and develop more robust countermeasures. The dataset undergoes preprocessing, such as removing duplicates, checking missing values, label encoding, k-means clustering, and min-max scaling to transform features suitable for further tasks. Then we split the data into train and test sets for training and evaluating the proposed method. After that we designed a K-means-based stacked model using three classifiers, such as RF, KNN and XGBoost. We trained the model using 80% of the data, and the remaining was kept separate for evaluation purposes. Figure 1 shows the proposed structure for attack detection in healthcare domain.

Workflow

We provided the step-by-step description of the workflow and, in Fig. 2, a flowchart that shows the interaction between preprocessing, clustering, and classification. The following sections provide an explanation of each step.

Preprocessing The IoHT data is preprocessed through removing duplicates, missing values, label encoding, and scaling.

Clustering We applied K-means clustering for feature engineering and selected the best features with $k=5$, generated cluster labels and appended them to the data, and then normalized the features suitable for further processing.

Feature enrichment The cluster-based features and labels are appended to the base feature set, enriching the input representation.

Base classifiers The base classifiers are employed, and the enriched features are fed into them. Each base classifier generates predictions using an enriched feature set.

Stacking The final model (stacked model) receives the predictions of the base classifiers and learns to combine them optimally for attack detection.

Data preprocessing

Preprocessing is a very crucial step in classification-related tasks to clean and transform the data into a suitable format for artificial intelligence models. We used techniques to prepare the data for healthcare attack detection. Some features contain non-numeric data, so we convert this data into a numeric format using `.to_numeric()` in Python. To ensure consistency, we set non-convertible data to NaN and remove them using `dropna()`. Furthermore, we use `astype('category').cat.codes` to transform the categorical columns 'Protocol', 'Source', and 'Destination' into numerical values. Label encoding provides the advantage of incorporating numerical features while maintaining their original categorical relationships. LabelEncoder transforms the target variables 'Type of attack' and 'Type' into a numerical format, assigning an integer to each unique attack type. Additionally, we apply a min-max scaler to normalise the input features and bring them into the same scale. This procedure prevents features with larger ranges from dominating others during model training. Preprocessing steps for attack detection using IoHT data is shown in Algorithm 1.

Kmeans clustering

This study enriched the data using K-means clustering. It is a method of grouping data points into specific groups or clusters based on their similarity. In this case, it is not used for predictive purposes, but to enrich the data set by identifying patterns or clusters within it. The process begins by selecting a set of groups (case 5). Five initial centers are then selected and sorted until each data point falls into the cluster closest to the data

Ref.	Methodology	Dataset	Results	Limitations
18	DNN	ECU-IoHT	The proposed system achieved 99.3%, 96.8 and 90.3%, precision, Recall and F1 Score, %, respectively.	The authors did not employed XAI techniques for explainability and no feature engineering to save the computational resources.
19	EFL-LSTM	ECU-IoHT	The author achieved 97.16 accuracy, 79.33 recall, 93.96 precision, 82.37 f1 score and 83.63 MCC.	Lack of interpretability, computational overhead and resources
21	CLSTM	ECU-IoHT	The author achieved 98.5% accuracy.	Lack of interpretability may limit trust in clinical settings.
23	RNN	ECU-IoHT	The author achieved 98.9% accuracy.	RNN is computationally expensive and authors did not employed XAI techniques.
15	DAG-LSTM	ECU-IoHT	The author achieved 92.04% averagetesting accuracy.	Overfitting issues, lack of Explainability and Trustworthiness, scalability
27	MD-LSTM	ECU-IoHT	The author achieved 99.42% accuracy for signature detection.	Low accuracy and computational overhead
28	ML	WUST-EHMS 2020	The author achieved 97.91% accuracy.	Lack of defense against AML attacks, lack of interpretability
29	CNN-LSTM	CIC-IDS2017	The author achieved 97.63% accuracy	Low accuracy, no feature engineering

Table 1. Summary of literature review.

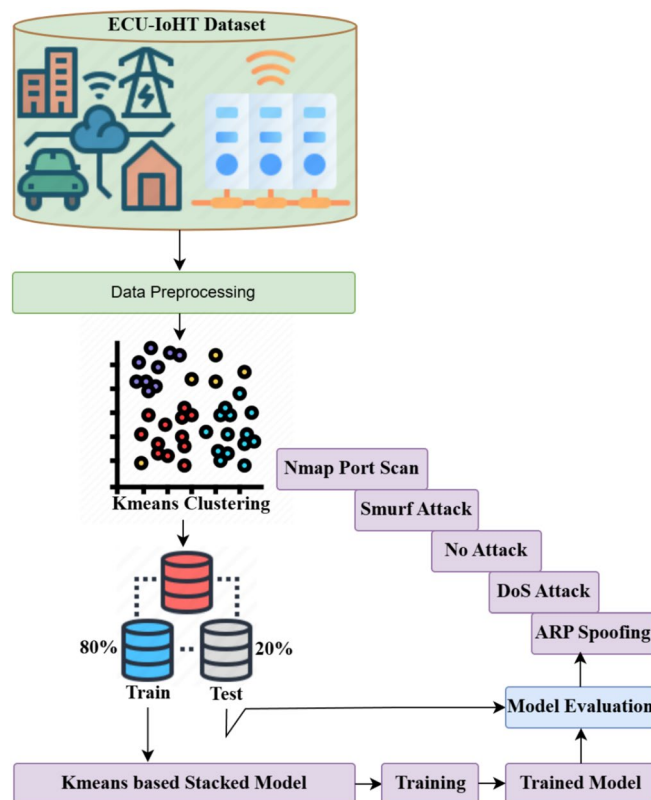


Fig. 1. Proposed structure for attack detection in healthcare domain. The raw dataset is collected, preprocessing is performed to transform the data into model format, clustering is applied, the data is split, the stacked model is designed, the model is trained, and then the trained model is tested using 20% test data.

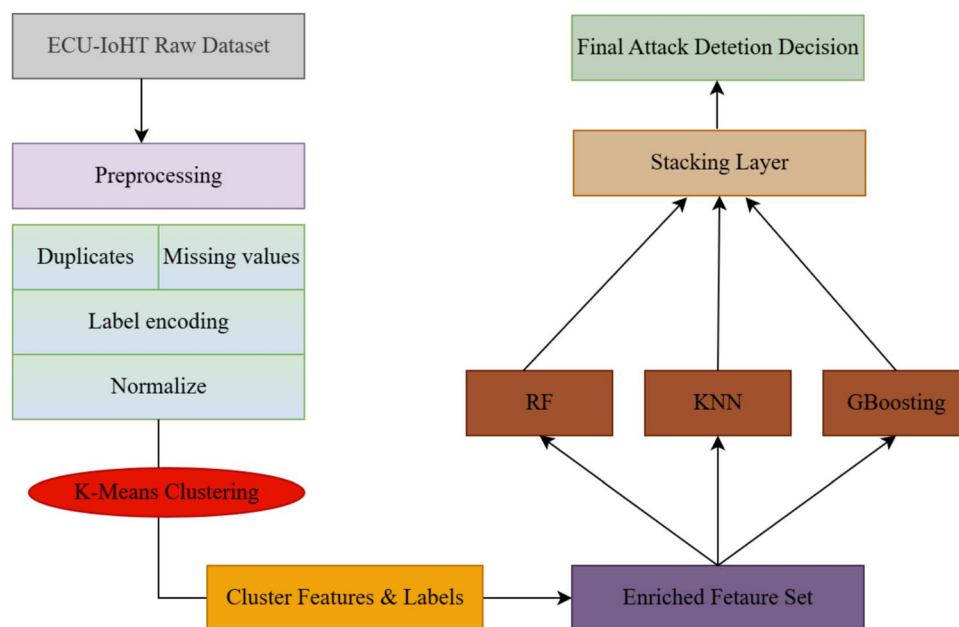


Fig. 2. Flow-Diagram for IoHT based Stacked Model. The ECU-IoHT raw dataset is preprocessed by removing missing values, duplicates, and label encoding; converting non-numeric features; and performing normalization. Thereafter, K-means clustering grouped similar data points together into clusters based on their features. Then enriched features are fed into base classifiers, and the predictions of each classifier are then combined with a stacking layer.

point. This ensures that objects in a cluster are more similar than other groups. The goal is to make each cluster as sparse as possible, which means that the members of the cluster are as close to each other in terms of data as possible. Here, the grouping is done in parameters. After grouping, each data point is assigned a number (0 to 4) that identifies its grouping, and this label is stored in a new column. These models are often useful for identifying unusual or anomalous activity in tasks such as monitoring network traffic or detecting unusual behavior. Identifying and using natural clusters in your data in conjunction with traditional data columns can improve accuracy and understandability, especially when dealing with complex or unusual data such as network monitoring or system logs³⁰.

```
Require: Raw Dataset  $D$  with Attributes: No., Time, Protocol, Length, Source,
          Destination, Info, Type, Type of attack
Ensure:  $X_{\text{scaled}}$  and  $Y$ 
1: Load  $D$ 
2: // Step 1: Convert non-numeric features
3: Convert Time, Length to numeric types  $\rightarrow Ti, Le$ 
4: Handle invalid entries with coercion
5: Drop all rows containing missing values  $\rightarrow D_{\text{cleaned}}$ 
6: // Step 2: Encode categorical attributes
7: Encode Protocol  $\rightarrow Pr_{\text{code}}$ 
8: Encode Source  $\rightarrow So_{\text{code}}$ 
9: Encode Destination  $\rightarrow D_{\text{code}}$ 
10: // Step 3: Encode target label
11: Apply label encoding on Type of attack  $\rightarrow Y$ 
12: // Step 4: KMeans clustering for feature engineering
13:  $X_{\text{base}} = \{Ti, Le, Pr_{\text{code}}, So_{\text{code}}, D_{\text{code}}\}$ 
14: Apply KMeans clustering with  $k = 5$  on  $X_{\text{base}}$ 
15: Generate cluster labels  $C$  and append to  $X_{\text{base}} \rightarrow X_{\text{raw}}$ 
16: // Step 5: Normalize features
17: Apply MinMaxScaler on  $X_{\text{raw}}$  to get  $X_{\text{scaled}}$ 
18: return  $X_{\text{scaled}}, Y$ 
```

Algorithm 1. Preprocessing steps for attack detection using IoHT data

Data splitting

To evaluate the performance and classification of a model, we split the data. It involves splitting the data into two different data sets: one to train the model and another to test its ability to predict unseen data. In this study, after preprocessing and feature set, the entire data set is split into a 80% train and 20% test set using train_test_split(). The $X_{\text{train}}, y_{\text{train}}$ is used to train the model and $X_{\text{test}}, y_{\text{test}}$ serves as a representative of real-world data that the model has never seen before. This evaluation is done to understand how well the trained model classifies the observed data. The data splitting values are shown in Table 2.

Stacked model

Stacked modeling is a sophisticated machine learning method that integrates several learning models to enhance the precision of predictions and classifications. Stacked models are different from typical models since they use many layers instead of only one. The base learner is the initial layer. It is made up of several classes, each of which makes its own predictions. A second layer model uses these predictions to figure out the best method to integrate the base learner’s findings to produce a final choice. In our case, we use the RF, KNN, and GBoosting classifiers for stacking. In the first step, we set three base learners and tune the parameters with a grid search algorithm. Then we applied the best grid search parameters for the three classifiers and designed a meta model

Attacks	Label	Training	Testing	Total
ARP spoofing	0	1915	444	2359
No Attack	3	18783	4670	23453
Nmap port scan	2	5441	1395	6836
Smurf attack	4	62320	15600	77920
DoS attack	1	506	133	639
Total		88965	22242	111207

Table 2. Slitting of data into train and test.

with a random forest classifier. In IoHT settings, stacking modules are very important for keeping track of the shape and interactions of network traffic data. There is something different about each of these algorithms. RFs are accurate and can take raw data and turn it into categorization decision trees. KNNs are easy-to-use, event-driven frameworks that perform well for making decisions and producing predictions. These student baseline files are not used for coding directly; instead, they are used as a guide for students. This strategy lets machine learning algorithms pick up on both low-level and high-level aspects of the input, which makes categorization more accurate. Proposed Stacked model for attack detection in IoHT is shown in Fig. 3.

The reason for having numerous layers in IOHT-based intrusion detection is because the data is typically complicated, nonlinear, and noisy. In this kind of setting, different kinds of assaults might show up in different kinds of data. K-Means clustering produces new features, or “clusters,” by putting unstructured sequence data into feature space. This novel method keeps the common clusters in the data, which makes the sequence model more accurate. MinMaxScaler scales all the characteristics so that algorithms may be compared accurately. This is especially crucial for complicated models like KNN. Kmeans based Explainable Stacking Model is visualized in Algorithm 2.

```

1: StackingClassifier  $\rightarrow SC$ 
2: Sample  $\rightarrow samp$ 
3: Base learner  $\rightarrow BL$ 
4: cross validation  $\rightarrow cv$ 
Require: Cleaned Dataset  $CD$ 
Ensure: Prediction and Explainability
5: Load  $CD$ 
6: Splitting  $CD \leftarrow$  Train, Test
7: Step 1: Define BL
8: Define base models:
9:    $M_1 \leftarrow$  RF Classifier
10:   $M_2 \leftarrow$  K-NNClassifier
11:   $M_3 \leftarrow$  GBoosting Classifier
12: Step 2: Define Meta Model
13: Meta-model  $M_{mt} \leftarrow$  RF Classifier
14: Step 3: Design Stacking Model
15: Stacked Model  $S \leftarrow SC(M_1, M_2, M_3, M_{mt}, cv)$ 
16: Training  $S$  using train data:  $S.fit(X_{train}, y_{train})$ 
17: Step 4: Predictions
18: Predict L:  $\hat{y} \leftarrow S.predict(X_{test})$ 
19: Predict prob:  $\hat{p} \leftarrow S.predict_proba(X_{test})$ 
20: Step 5: Evaluation
21: Calculate Metrics:
22:   Accuracy, Precision, Recall, F1-Score
23:   MCC, Kappa, Log Loss
24:   ROC AUC
25: Step 6: Explainability via SHAP and LIME
26: (a) Explainability via SHAP
27: Select sample:  $X_{sample} \leftarrow X_{test}[no.of.samples]$ 
28: Initialize SHAP:  $Explanations \leftarrow shap.Explanations(S.predict_proba, X_{train})$ 
29: Calculate SHAP values:  $shap\_values \leftarrow Explanations(X_{sample})$ 
30: for each class  $c$  in all classes do
31:   Generate plot for class  $c$ 
32: end for
33: (b) Explainability via LIME
34: Initialize LIME Explanations with  $X_{train}$ 
35: Select instance  $x_i \leftarrow X_{test}[i]$ 
36: Explain  $x_i$  via LIME:
37:  $lime\_Explanations \leftarrow LIME.Explanations(x_i, S.predict_proba)$ 
38: Display Explanation

```

Algorithm 2. Kmeans based Explainable Stacking Model

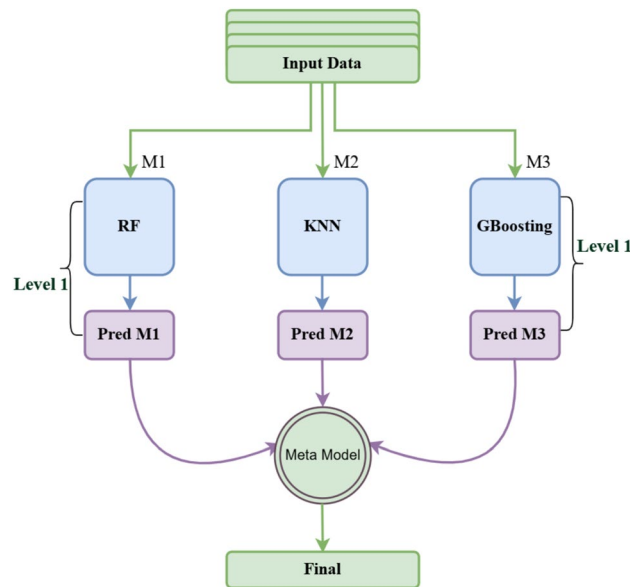


Fig. 3. Proposed stacked model for attack detection in IoHT: here, three base-tuned classifiers are designed, such as RF, KNN, and GBoosting at level 1; at level 2, the predictions of each classifier are combined to the meta-model (stacking model) to make the final decision.

Model training and evaluation

We used three different classifiers, RF, KNN, and GBoosting, to develop the stacking model and then do training using the training data. After training the models, their predictions are passed to another model called the meta-learner. The meta-learner looks at the outputs of the base models and learns how to combine them to make a final decision. Performance measures, which are sometimes called metrics, are necessary for many types of classification tasks. Using metrics, machine learning compares the predictions made by the trained model to the real data in the testing data set. The precise forecast rate refers to the number of observations that accurately predicted the outcome. However, using accuracy alone to judge how useful a model is not a good idea. When the data are balanced and there are about the same numbers of FPs and FNs, accuracy is key.

In the context of imbalanced datasets such as ECU-IoHT, a purely rule-based approach can be successful, as rule-based results often dominate risk outcomes. It is important to monitor how the model minimizes false positives, which is especially important in clinical settings to avoid unintended consequences. It is important to remember that the main concept is not to overestimate a person's ability to retain a realistic experience. Using these measures allows for a clearer image of the strengths and weaknesses of these algorithms. By aligning the study with advances in cybersecurity research, this solution will ultimately become more relevant and suitable for real IoT applications.

The accuracy alone can be misleading, in the case where the datasets are imbalanced where some class has more samples than others. Precision avoids the false positives make it effective and important in healthcare. The recall measure the ability to detect the actual attacks, which is critical since missed threats can endanger patient safety. The F1-score, also called harmonic mean of precision and recall, provides a balanced representation. This makes the findings more useful for anyone trying to understand or improve attack detection in real-world IoHT systems.

$$Accuracy = \frac{TP + TN}{TP + FN + FP + TN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 - Score = 2 \times \frac{PRE \times REC}{PRE + REC} \quad (4)$$

SHAP and LIME

SHAP and LIME are included in this study to ensure the clarity and interpretability of the proposed framework. When the model is trained, the features are used to inform the classifier's decision-making process and evaluate the relative importance of each feature. SHAP is used to assess feature importance values by assigning a shapley value to each input feature. This allows us to measure the global size of the nodes in the dataset, providing insight into which network traffic characteristics determine the most malicious and non-malicious activity patterns. On

the other hand, LIME is used to generate local solutions by fitting fitness values to individual observations and applying a random number of steps. This has helped us understand the behavior of the model for specific cases. Both methods are used not only to verify model accuracy but also to ensure integration and domain knowledge in IoHT security. By combining global interpretation with local interpretation, this framework increases patient confidence and facilitates effective assessment in healthcare settings.

Results and discussion

This section presents a comprehensive evaluation of the experimental results using deep learning models and the proposed explainable stacked model. In this, first, we explore the information from the data set, data analysis, and anomalies, and then perform deep models for the detection of cyberattacks in IoHT. Subsequently, we conducted experiments using the stacked model. After the comparison of the proposed and the other models, we explain the model decisions using XAI techniques SHAP and LIME. The subsections provide the results and discussion.

Dataset information

This study tests the proposed technique using the ECU-IoHT dataset³¹, a novel dataset in the healthcare sector. By mimicking various assaults, this dataset created in an IoHT environment reveals several vulnerabilities. Healthcare security personnel can use it as a reference to examine attack tendencies and enhance their defences. The data collection contains 111,207 instances that range from typical network traffic to intrusions in the healthcare industry. The information includes seven aspects of the network data: time, source, location, protocol, duration, type, and specific types of network assaults. Among the 111,207 samples included in the dataset, 23,453 exhibit typical network activity, whereas the remaining 87,754 exhibit intrusion events. These assaults can take four forms: smurf attacks, denial of service attacks, ARP spoofing, and nmap port scan. Figure 4 provides a comprehensive breakdown of the numbers for various types of assaults, including typical ones.

The dataset was developed through a controlled IoT testbed that simulates a healthcare environment. White hat penetration testing techniques were used to detect vulnerabilities and realistic cyberattacks. This type of controlled environment allowed researchers to collect network traffic and system logs while maintaining accurate labeling, which would not be possible in a real hospital due to privacy constraints. The collected dataset has limitations in real-world applications such as the testbed may not fully capture the heterogeneity of devices, network conditions, and unpredictable workflows of a real hospital, and its limited attack scenarios may not reflect rare or zero-day attacks.

Data analysis

Data analysis is very important step in machine learning to understand the data structure, its patterns and anomalies. Some features are plotted using the pair-plot for binary class as shown in Fig. 5. Here feature are visualized to show the relationship between the classes. Also, in Fig. 6 KDE plot for the distribution of features across binary class traffic type is shown. The script filters the dataset and then creates a series of KDE plots using sns.kdeplot, with the space below the plot blank to improve visual clarity. Each poster is labeled based on its corresponding category and traffic type. Logarithmic scaling is applied to the x-axis using plt.xscale('log'), which is especially useful when the time values vary greatly or exhibit a skewed distribution. The labels include a title, a bullet list, and a legend explaining what each label means. This concept helps to show different traffic accidents at different times and to distinguish between different types of attacks. Their occurrence in clusters over time is a useful insight for analyzing survey data or predicting trends in safety issues. In Fig. 7, we also show some features and their correlation using pair-plot for Multi-attacks such as Smurf, Nmap port scan, ARP spoofing, DoS attack. KDE plot of feature distribution for different traffic types is shown in Fig. 8.

Figure 9 illustrate the attack type over time (A) for binary, (B) for Multi-attacks and (C) for traffic with anomaly detection. The proposed framework analyzes network traffic data over time to identify patterns, and anomalies that may indicate security vulnerabilities or changes in user behavior. For time series analysis, we first create a series that needs to be replicated and then convert the data into a time series. Linear regression

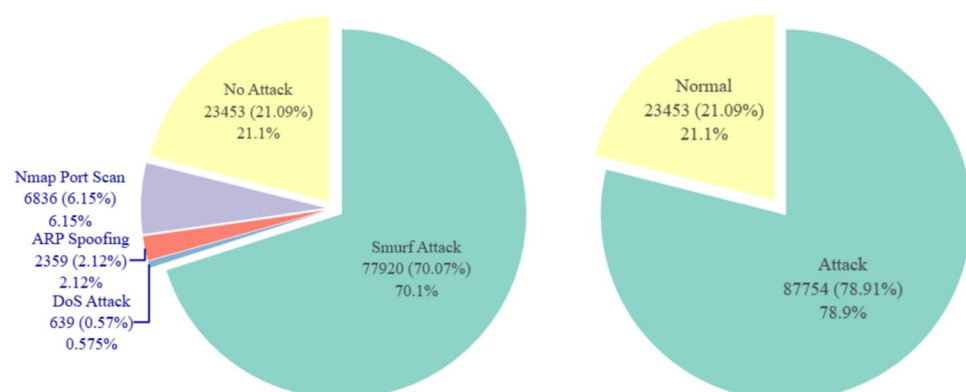


Fig. 4. The number of attacks in ECU-IoHT Dataset for binary and Multi-attacks.

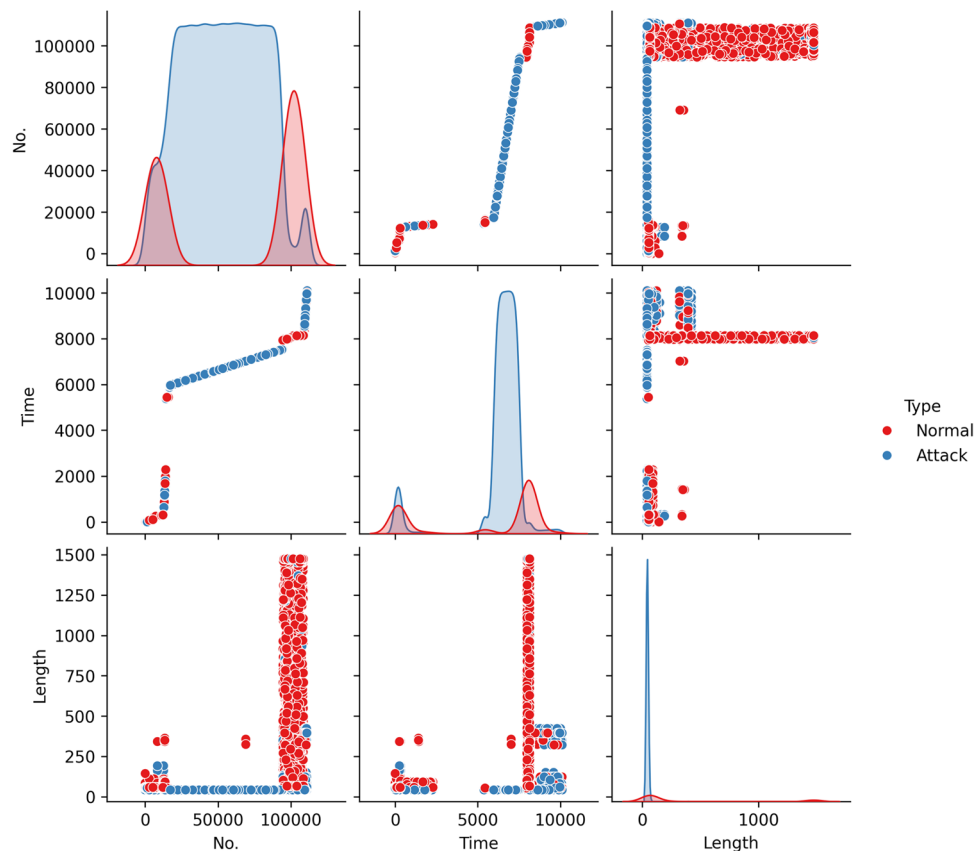


Fig. 5. Pair-plot for visualization of features for binary class.

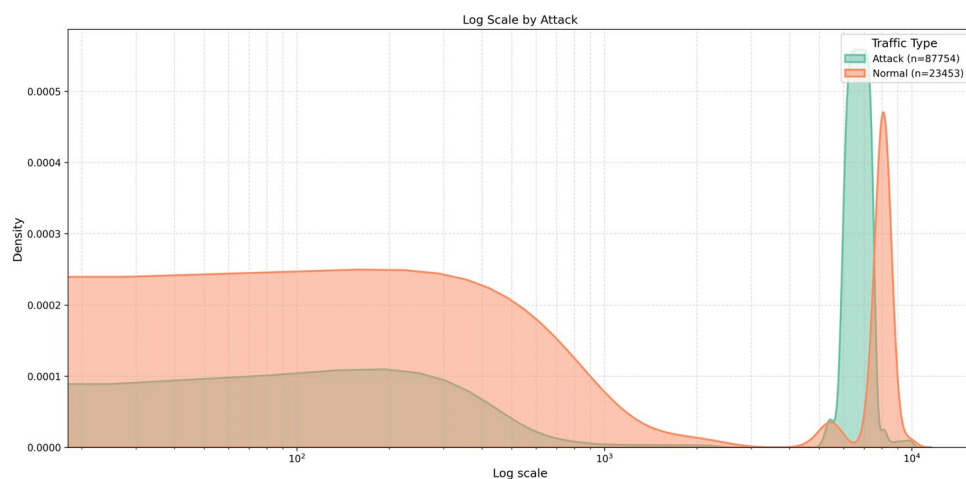


Fig. 6. KDE plot for the distribution of features across binary class traffic type.

was used to split the data into 10-second intervals to suppress short-term fluctuations and reveal global trends. Comparing speeds across time series and traffic patterns allows us to create traffic graphs that show how each category normal traffic or attack traffic evolves over time. It helps to identify normal or abnormal activity.

This graph allows us to further investigate the nature of specific errors for only the records associated with errors. By studying the development of individual attacks over time, we can identify their trends and timing, thus facilitating prevention and response strategies. To do so, we conducted a post hoc analysis using the Z - score method. Any period with a value greater than 3 or less than -3 was classified as an outlier, and this was reported in the spreadsheet. The method quickly detects anomalies that could indicate previously unknown or hidden attack attempts. Overall, this study provides a useful framework for understanding and analyzing different network dynamics using time series, distributions, and statistical principles.

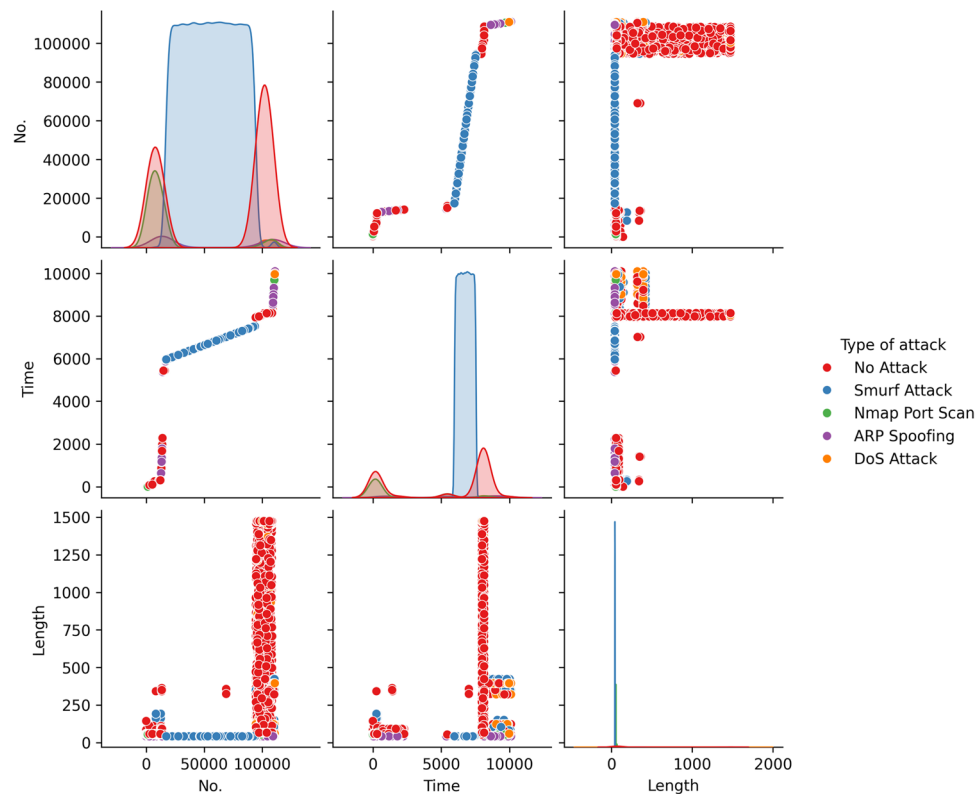


Fig. 7. Pair-plot for visualization of features for multi-attacks.

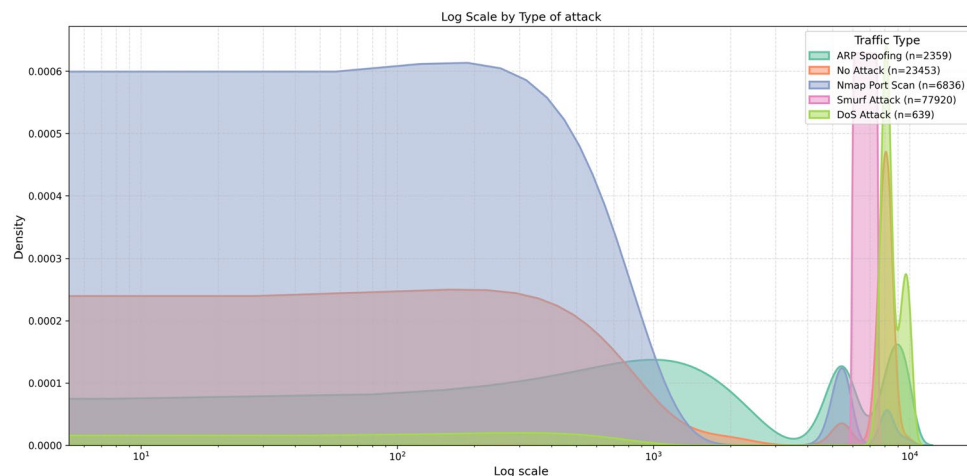


Fig. 8. KDE plot for the distribution of features across multi-attacks.

Cyberattacks analysis in IoHT using the proposed model

The proposed stacked model performance for cyberattack detection in IoHT is shown in Table 3. The proposed stack achieved 100% performance for two attacks, such as ARP spoofing and Smurf attacks. For the Nmap port attack, the model achieved 99.12% precision, 99.95% AUC, 97.75% F1 score and 96.42% recall score. For the DoS attack, the model achieved 78.57% precision, 71.84% F1, 66.17% precision and 99.85% AUC score.

In addition, along with key metrics, we also performed experiments using other metrics, such as MCC, kappa and log loss. Figure 10 presents the performance of the proposed model. The model achieved 98.72% MCC and kappa and 0.0221 log loss. MCC is a performance metric may measure how well machine learning multiclass classifications hold up. It is often considered a fair metric to use when the class sizes are significantly different. It encompasses both real and deceptive positives, as well as negatives. The MCC is fundamentally a correlation coefficient value between -1 and $+1$. A coefficient of 1 means a perfect forecast, and -1 means a entirely wrong forecast.

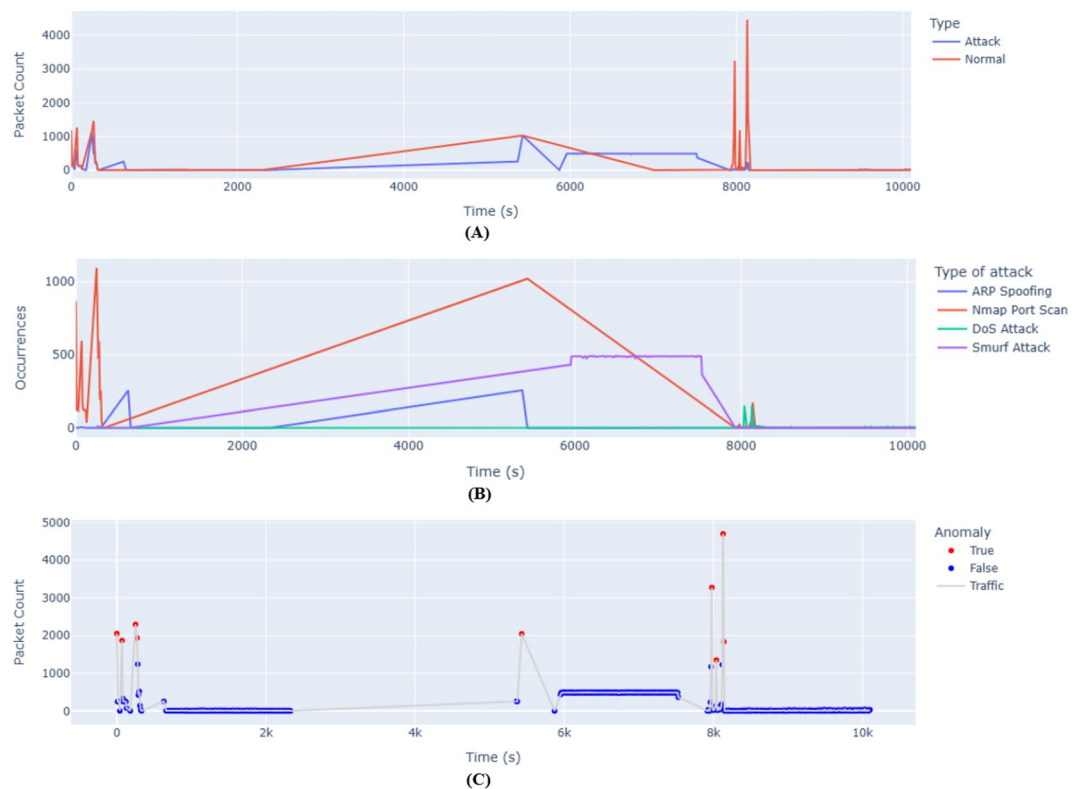


Fig. 9. Attack type over time (A) for binary, (B) for Multi-attacks and (C) Traffic with anomaly detection.

Attacks	Precision	Recall	F1 score	AUC
ARP spoofing	1.0000	1.0000	1.0000	1.0000
No attack	0.9799	0.9923	0.9861	0.9996
Nmap port scan	0.9912	0.9642	0.9775	0.9995
Smurf attack	1.0000	1.0000	1.0000	1.0000
DoS attack	0.7857	0.6617	0.7184	0.9985

Table 3. Proposed stacked model performance for cyberattack detection in IoHT.

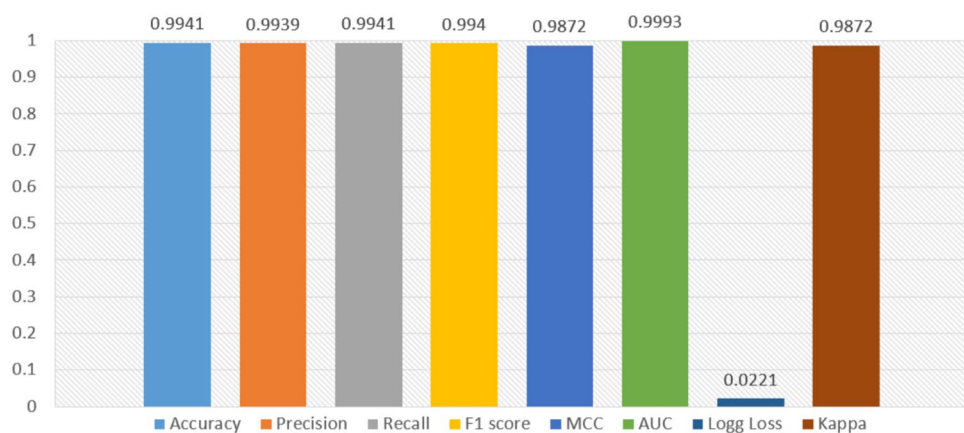


Fig. 10. Overall performance in detection of cyberattacks in IoHT for multi-attacks.

MCC is calculated as:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (5)$$

Log Loss is a logarithmic change to the likelihood function that is mostly used to check how well probabilistic classifiers work. Log Loss is different from other measures like accuracy since it takes into consideration how uncertain predictions are by giving models a bigger penalty for confidently wrong predictions. In Eq. 2, N represents total number of samples, y_i true label and p_i represents the predicted probability.

$$\text{LogLoss} = -\frac{1}{N} \sum_{i=1}^N (y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)) \quad (6)$$

Learning curve and ROC

When evaluating algorithms that learn from a training dataset, learning curves are a common diagnostic method in machine learning. After each training update, the proposed model is tested holdout validation datasets, and graphs of the measured performance are created in Fig. 11 to show the learning curves. We can determine whether a model is well-fit, overfit, or underfit by analysing its performance learning curves on the validation and training datasets. Determine if the train or validation datasets do not adequately reflect the problem domain, look at the learning curves of the model's performance.

Comparison of proposed stacked model performance with other models

The proposed stacked model is contrasted with other models that use the most powerful deep learning methods as presented in Table 4 and aim to achieve the same goal in this section. The comparative analysis use the same data to show how well each model performs in comparison. The RNN model achieved 91.31% accuracy, 89.67% F1 score, 98.08% AUC, 80.90% MCC and 0.1872 log loss. The LSTM achieved some better results as compared

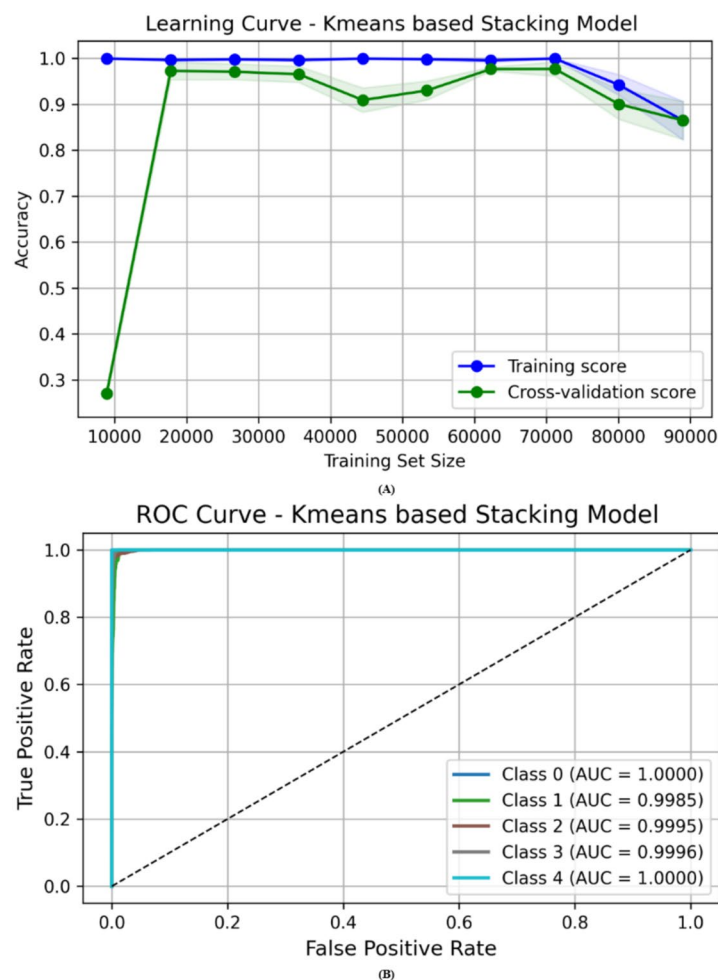


Fig. 11. Learning curve and ROC for multi-attacks.

Performance	RNN	LSTM	GRU	CNN	Proposed
Accuracy	0.9131	0.9334	0.9324	0.9902	0.9941
Precision (weighted)	0.8950	0.9227	0.8861	0.9906	0.9939
Recall (weighted)	0.9131	0.9334	0.9324	0.9902	0.9941
F1 score (weighted)	0.8967	0.9187	0.9044	0.9893	0.9940
AUC	0.9808	0.9898	0.9886	0.9973	0.9993
MCC	0.8090	0.8570	0.8582	0.9788	0.9872
Kappa	0.8055	0.8527	0.8497	0.9786	0.9872
Log loss	0.1872	0.1256	0.1313	0.0490	0.0221

Table 4. Comparison of proposed stacked model performance with other deep learning models for cyberattack detection in IoHT.

Model	T test	P value	Significant
Proposed vs RNN	18.1014	0.0004	Yes
Proposed vs LSTM	18.0123	0.0004	Yes
Proposed vs CNN	13.7521	0.0008	Yes
Proposed vs GRU	7.0919	0.0058	Yes

Table 5. Statistical t test.

Model	Training time (seconds)	Scalability
RNN	12 per epoch	Low
CNN	6 per epoch	High
LSTM	8 per epoch	Medium
GRU	10 per epoch	Medium
Proposed	640 seconds	Medium

Table 6. Scalability for DL and the proposed model.

with RNN and GRU and achieved 93.34% accuracy, 98.98% AUC, and 85.27% kappa score, respectively. The GRU achieved 0.1313 log loss and 93.24% recall score. The CNN model achieved 99.02% accuracy, 98.93% AUC, 97.88% MCC and 0.0490 log loss. In deep learning, RNN performed less accurately than LSTM and GRU, but the CNN model achieved good results in terms of all metrics. The proposed model achieved 99.41% accuracy, 99.93% AUC, 98.72% MCC and 0.0221 log loss. Overall, results ensure that the proposed model achieved outclass results for attack detection.

Statistical test

To validate that the observed improvement in detection accuracy is statistically significant, we perform t-tests and compare our proposed framework with baseline DL methods. The results show that the performance of our method is statistically significant compared to existing methods ($p < 0.05$), indicating that the improvement is not limited to the robustness of the performance of the proposed method in detecting discrepancies and IoT attacks. Table 5 demonstrate the statistical test.

Model scalability

The evaluation of the training and execution time reduction of the proposed classification model over CNN, RNN, LSTM, GRU shows an important trade-off between accuracy and efficiency for IOHT applications as shown in Table 6. CNN showed faster training time with ease of use, making it suitable for attack detection. RNN, although moderately efficient, suffers from slow processing, which reduces the ability to simulate real-time clinical environments. LSTM and GRU provide performance improvements by generating features, but their high training time and computational requirements limit them to lightweight IOHT tools without optimization. The proposed model maintained competitive scalability by enriching feature representation through K-Means clustering and leveraging multiple classifiers for robustness. This balance between accuracy and scalability suggests that the proposed framework is suitable for real-world IoHT deployment, provided further optimization is performed to minimize energy consumption and inference latency.

Explainable via SHAP and LIME

Explainable AI refers to a set of methods that help to explain how a particular AI model makes a decision. With the ongoing emergence of new and more advanced approaches, this relatively new area of AI has demonstrated

tremendous promise. Some of the most well-known XAI approaches include SHAP and LIME. For smaller models, LIME's localized interpretability is perfect for deciphering specific predictions. Any kind or complexity of machine learning model can be used with LIME because it is a model-agnostic interpretability tool. By introducing noise into the input data in order to provide local approximations, it may be used with a wide variety of models, from basic linear regressions to complicated neural networks. SHAP explanation using the proposed model for multi-class attacks are Visualization in Fig. 12, for LIME in Figs. 13 and 14.

To improve model performance, we used SHAP and LIME in the Windows framework, which provide both global and local annotations. SHAP was used to evaluate the effectiveness of each feature in the database, highlighting the most important network traffic paths for detecting attacks. LIME was used to make individual predictions to explain why certain events were classified as outliers or specific features. These annotation techniques help healthcare professionals understand the reasoning behind model performance, thereby increasing confidence in automated prediction systems. For example, SHAP analysis showed that features such as cluster frequency and connection frequency are more important for detecting attacks. LIME annotations helped doctors and security professionals analyze early warnings and identify key trends, enabling them to make important decisions such as monitoring patients or protecting sensitive medical equipment.

SHAP and LIME explanations show that features such as source, destination, and protocol type are very useful for distinguishing legitimate and malicious IoT traffic. Unusual traffic patterns and unusual packet characteristics are indicators of network attacks in healthcare IoT environments, and these results are consistent with domain knowledge. For example, large clusters or unusual clusters contribute significantly to attack classification, which limits the accuracy of the model's relationship to real-world sample situations. By combining these explanatory tools, our framework not only improves transparency but also provides actionable insights that healthcare professionals can rely on to make critical decisions.

Experiments with additional WUSTL EHMS dataset

Hady et al.³² generated WUSTL-EHMS dataset in real-time using an enhanced healthcare monitoring system testbed. This testbed collects information on patients' biometrics as well as the volume of data sent across a network, in response to the lack of datasets that aggregate many biometric categories. The EHMS testbed includes medical sensors, a gateway, a network, and control and visual systems. The Intrusion Detection System (IDS) is responsible for identifying anomalies, tracking network activity in real-time, and gathering patient

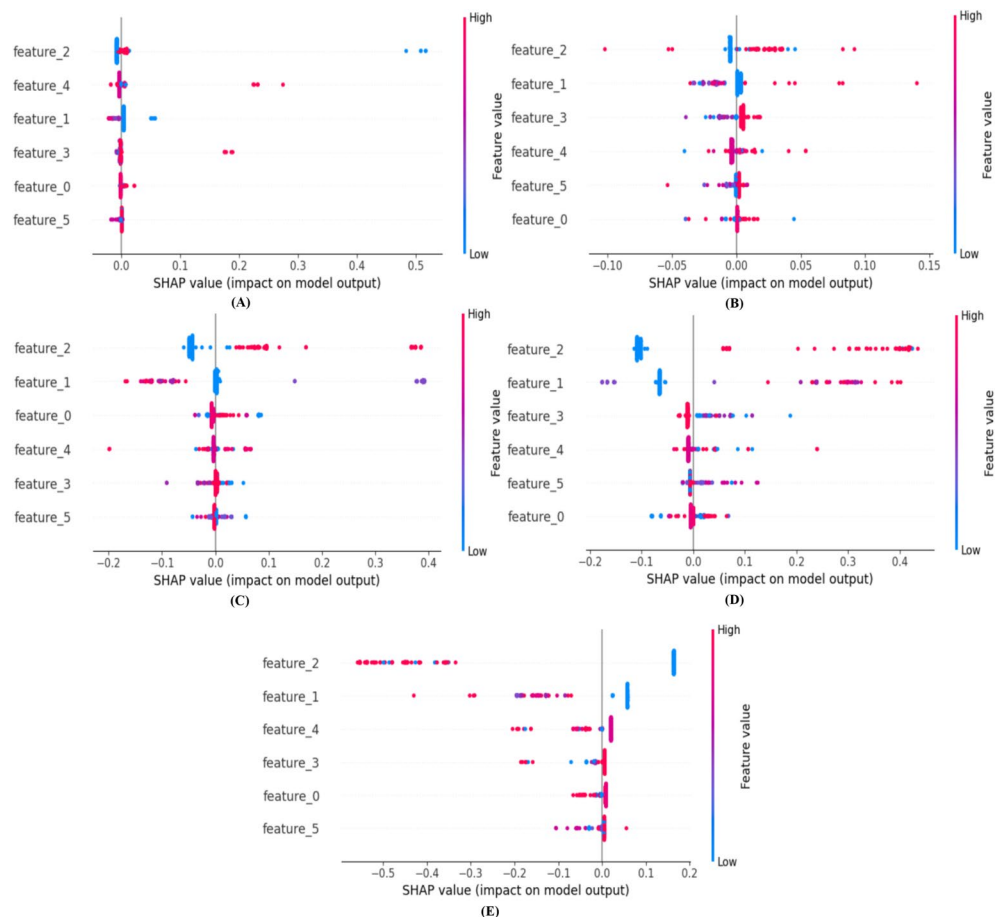


Fig. 12. Visualization of SHAP explanation using the proposed model for multi-class attacks: (A) for attack 0, (B) for attack 1, (C) for attack 2, (D) for attack 3, and (E) for attack 4.

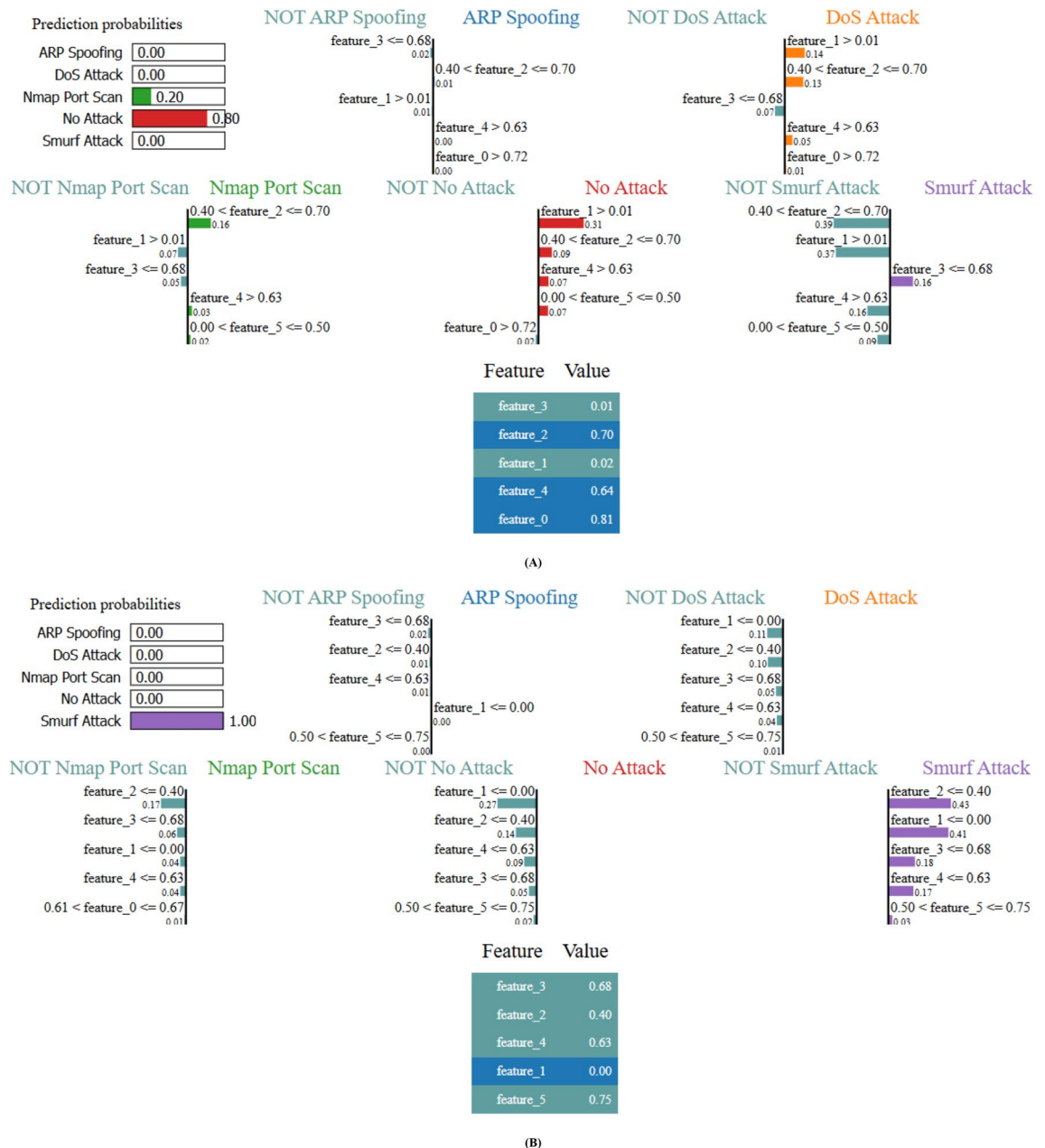


Fig. 13. LIME explanation using the proposed model with highest probabilities: (A) No attack, (B) Smurf attack.

biometric information. Network traffic statistics and the patient's biometric information were recorded using an application named ARGUS. The dataset contains 44 features, including 14,272 normal instances and 2,046 spoofing attacks. Table 7 shows the performance of proposed model on WUSTL EHMS dataset. Our model achieved superior performance in attack detection with a 0% error rate.

Comprehensive comparison of proposed stacked model with state-of-the-art methods

A comparison is made between the performance of the proposed strategy and those of previous research that utilized the same dataset in order to search for potential attacks in the IoHT. We examine the accuracy, precision, recall, F1 score, and AUC of the recommended approach in order to evaluate how well it performed. The comparison is shown in Table 8. The study¹⁶ used the ECU-IoHT dataset for the detection of IoHT attacks with the MobileNet-CNN hybrid model and achieved 97.1% recall and 98.4% F1 score. The study¹⁸ achieved 96.8% recall and 90.3% F1 score utilising the same dataset with the DNN model. We compared only those studies that used the same dataset for better comparison. Another research¹⁹ used a federated learning-based LSTM and achieved 97.1% accuracy and an 83.3% MCC score. This two-pronged strategy not only improves detection precision but also assures the continuous security of private medical information. Authors in²¹ employed a one-

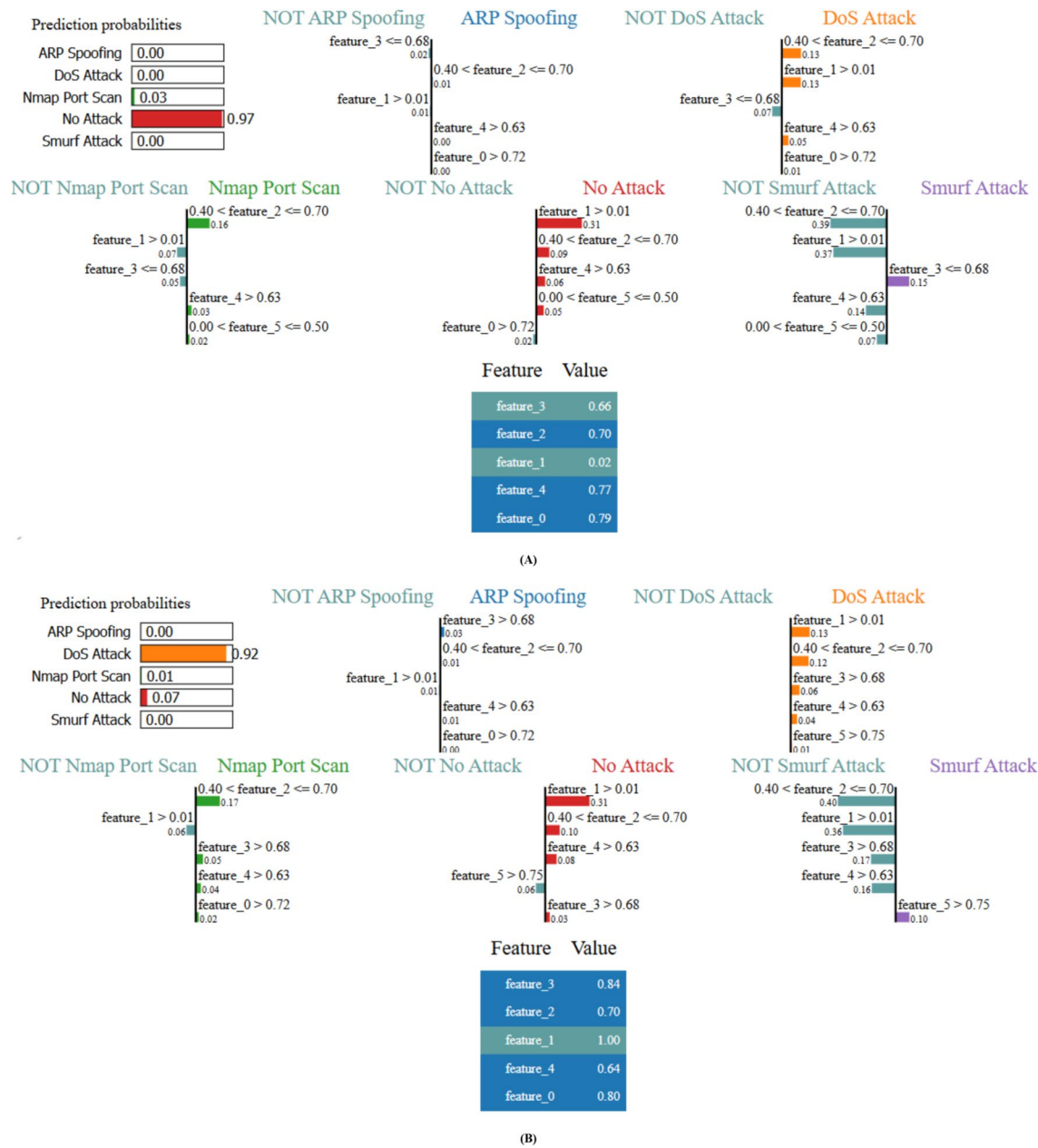


Fig. 14. LIME explanation using the proposed model with highest probabilities: (A) No attack, (B) DoS attack.

Attacks	Precision	Recall	F1 score	AUC
Normal	1.0000	1.0000	1.0000	1.0000
Spoofing	1.0000	1.0000	1.0000	1.0000
Accuracy				1.0000
Macro avg	1.0000	1.0000	1.0000	–
Weighted avg	1.0000	1.0000	1.0000	–

Table 7. Performance of proposed model on WUSTL EHMS dataset.

dimensional CLSTM model for the detection of attacks in IoHT and achieved 98.5% accuracy. Rodge et al.²³ also employed a deep learning model and detected attacks with 98.9% accuracy. Similarly, using the same dataset, the DAG-LSTM model achieved very poor accuracy on test data in study¹⁵. To record and convey layer-by-layer dependencies on time, the writers devised the Memory Feedback Transformer (MF-Transformer). Memory Feedback LSTM was developed into the Transformer architecture of this device. The remarkable performance

SOTA studies	Dataset	Methodology	Performance
16	ECU-IoHT	MobileNet-CNN	The author achieved 99.1% accuracy, a precision of 99.00%, a recall of 97.1%, and an F1 score of 98.4%.
18	ECU-IoHT	DNN	The proposed system achieved 99.3%, 96.8 and 90.3%, precision, Recall and F1 Score, %, respectively.
19	ECU-IoHT	EFL-LSTM	The author achieved 97.16 accuracy, 79.33 recall, 93.96 precision, 82.37 f1 score and 83.63 MCC.
21	ECU-IoHT	CLSTM	The author achieved 98.5% accuracy.
23	ECU-IoHT	RNN	The author achieved 98.9% accuracy.
15	ECU-IoHT	DAG-LSTM	The author achieved 92.04% average testing accuracy.
27	ECU-IoHT	MF-LSTM	The author achieved 99.42% accuracy for signature detection.
28	WUST-EHMS 2020	ML	The author achieved 97.91% accuracy.
29	CIC-IDS2017	CNN-LSTM	The author achieved 97.63% accuracy.
33	ECU-IoHT	Federated Learning	The author achieved 95.48% accuracy.
34	ECU-IoHT	DNN-FL	The author achieved 98.47% accuracy.
Proposed	ECU-IoHT	Stacked	The proposed model achieved 99.41% accuracy, 99.39% precision, 99.40 f1 score, 99.93 AUC, 98.72 MCC, and 0.0221 loss.

Table 8. Comprehensive comparison of proposed stacked model with state-of-the-art methods.

of the proposed MF-Transformer was demonstrated by its 99.42% accuracy rates in signature detection²⁷. An intrusion detection system tailored to the unique characteristics of IoMT medical equipment in order to keep them secure. The system used recursive feature reduction, advanced machine learning models, and the SHAP mechanism. Overall, the system achieved a 97.91% accuracy rating on the WUST-EHMS 2020 dataset²⁸. The goal study²⁹ was to provide a novel IDS called SafetyMed that could protect sequential and grid data from infiltration by combining LSTM networks with CNN. A 97.63% success percentage was attained. With a 95.48% success rate, the authors suggested using a privacy-preserving federated learning system to make IoHT data more secure³³. In another paper³⁴, authors presented a method for detecting security-related anomalies in IoHT traffic using a deep neural network with federated learning (DNN-FL). They looked at the proposal's detection performance based on metrics like accuracy and precision. Using the ECU-IoHT and wustl-ehms-2020 datasets, the suggested DNN-FL was verified. On the first dataset, the system correctly identified attacks at a rate of 91.40%; on the second dataset, it improved to 98.47%. Most studies utilised the deep learning approach and achieved balanced accuracy. The proposed model achieved over 99% performance using the same dataset for attack detection. The proposed model achieved 99.41% accuracy, 99.93% AUC and 98.72% MCC with 0.0221 log loss, ensuring that our approach performed superior to state-of-the-art (SOTA).

Applicability, privacy and limitations

Although our stacked architecture demonstrates robust performance on IOHT testbed, its application in a real-world environment poses additional challenges. Many IOHT devices operate on edge platforms with limited computing power, and memory, which can hinder real-time safety monitoring. Large-scale IOHT systems also generate unprecedented levels of network latency, which increases computational complexity and scalability. Feature selection and model tuning strategies, would reduce computational demands while maintaining acceptable classification performance. It is important to consider the trade-off between specificity and accuracy, to address these challenges. Edge devices perform initial analysis and send relevant data to central systems, can further improve analysis speed.

The proposed approach is based on detecting attacks in IOHT environments. However, it is also important to protect the integrity of sensitive health data. During model training, patient data such as medical records, device telemetry, and simulation parameters should be protected from loss or misuse to ensure robustness. Training can be performed on preprocessed data, while classification techniques can be applied to selected enriched dataset. Furthermore, we believe that machine learning models themselves can create vulnerabilities, such as model cloning attacks that expose unique features or conflicting detections. Controlling these risks is important, and combining security measures with proactive monitoring ensures that IOHT security solutions are safe for the system and the patient.

The proposed Smart Defence-IoHT framework showed excellent results against various attacks using the ECU-IOT and WUSTL EHMS datasets. However, the study has some limitations that are acknowledged.

First, the computational complexity of proposed model is somehow higher than that of individual classifiers. The training process involves multiple base learners and a meta-model, which can be resource-intensive, particularly when applied to large-scale or real-time healthcare environments. This complexity may hinder deployment on resource-constrained edge devices commonly used in IoHT systems.

Second, while K-means clustering improves the feature set, it has its own challenges. K-means clustering works by requiring the right number of clusters (k). If the models do not reflect significant patterns in the data, the new method will not be useful and may introduce bias into the model. The dataset is also very important. The dataset contains many common attacks, but it does not represent all breaches that could occur in real hospitals or medical monitoring systems.

Third, ECU-IoHT dataset was developed in a controlled IoHT testbed, it may not fully represent the scale, traffic diversity, and device heterogeneity of real-world hospital networks or wearable healthcare devices.

Therefore, while the dataset provides a valuable foundation for studying IoHT security, future efforts should focus on expanding to more complex, real-world scenarios to strengthen external validity.

Lastly, the proposed model achieved 100% performance for ARP spoofing and Smurf attacks but low performance for DoS attacks. In the future, we will employ more sophisticated advanced feature engineering and lightweight models and may test in real time.

Conclusion and future directions

This study presents an enhanced framework Smart Defence-IoHT for the detection of different cybersecurity attacks in IoHT. First we performed experiments using individual models then applied the proposed model to enhance the performance. This research presents analysis of the ECU-IoHT dataset, which is derived from an IoHT environment with distinct attacks that have many vulnerabilities. The healthcare security community hoped to use this dataset to better understand attack trends and build resilient defences. After data cleaning and performing K-means clustering on the dataset, we train the models. We evaluate the trained models using various key performance metrics. The experimental results indicate that the proposed model achieved overall best performance with 99.41% accuracy, 99.40% F1 score, 99.93% AUC, 98.72% kappa and 0.0221 log loss. For ARP spoofing and Smurf attacks, the proposed model achieved 100% performance for AUC, precision, recall and F1 scores. For the Nmap port scan attack, the model achieved 99.12% precision and 99.95% AUC. In addition, we validate our model effectiveness on another WUSTL EHMS dataset, where our model achieved 100% in precision, recall, and F1-score. This framework is ideal for medical things, enabling accurate monitoring and detection of security threats. By alerting healthcare workers to potential threats in a timely manner, it improves operational response and supports more efficient delivery of care.

The proposed stacked framework with k-means clustering provides superior performance in IOHT attack detection, its performance in real-time implementation still needs further validation, especially in terms of accuracy and execution time. Future work will explore extending security tools to enable timely release. In addition, qualitative and quantitative assessments are critical to ensure quality; thus, ensuring compliance with health data security regulations such as HIPAA and GDPR will lay the foundation for future improvements. We also aim to introduce privacy protection mechanisms such as FL to protect sensitive medical data. Ultimately, we hope to expand our work to real-time implementation testing, integration with healthcare platforms, cross-domain authentication, and improved security and patient insights.

Data availability

The datasets are publicly available online at: ECU-IoHT Dataset: Reference: Ahmed,M.,Byreddy,S.,Nutakki,A.,Sikos,L.F.,# &Haskell-Dowland,P. (2020).ECU-IoHT..<https://doi.org/10.25958/5f1f97b837aca> The WUSTL-EHMS-2020 Dataset: Link: <https://www.cse.wustl.edu/~jain/ehms/index.html> Reference: A.A.Hady,A .Ghubaish,T.Salman,D.UnalandR.Jain, "IntrusionDetectionSystemforHealthcareSystemsUsingMedicalandNetworkData: AComparisonStudy,"inIEEEAccess,vol.8,pp.106576-106584,2020,doi:10.1109/ ACCESS.2020.3000421.<https://ieeexplore.ieee.org/document/9109651>

Received: 7 July 2025; Accepted: 18 September 2025

Published online: 24 October 2025

References

1. Dimitrov, D. V. Medical internet of things and big data in healthcare. *Healthc. Inform. Res.* **22**(3), 156–163 (2016).
2. Yuehong, Y., Zeng, Y., Chen, X. & Fan, Y. The internet of things in healthcare: An overview. *J. Ind. Inf. Integr.* **1**, 3–13 (2016).
3. Jagadeeswari, V., Subramaniaswamy, V., Logesh, R. T. A. & Vijayakumar, V. A study on medical internet of things and big data in personalized healthcare system. *Health Inf. Sci. Syst.* **6**(1), 14 (2018).
4. Baseer, K. et al. Healthcare diagnostics with an adaptive deep learning model integrated with the internet of medical things (iomt) for predicting heart disease. *Biomed. Signal Process. Control* **92**, 105988 (2024).
5. AlZubi, A. A., Al-Maitah, M. & Alarifi, A. Cyber-attack detection in healthcare using cyber-physical system and machine learning techniques. *Soft. Comput.* **25**(18), 12319–12332 (2021).
6. Thomasian, N. M. & Adashi, E. Y. Cybersecurity in the internet of medical things. *Health Policy Technol.* **10**(3), 100549 (2021).
7. Stellios, I., Kotzanikolaou, P., Psarakis, M., Alcaraz, C. & Lopez, J. A survey of iot-enabled cyberattacks: Assessing attack paths to critical infrastructures and services. *IEEE Commun. Surv. Tutor.* **20**(4), 3453–3495 (2018).
8. Ibarra, J., Jahankhani, H. & Kendziarskyj, S. Cyber-physical attacks and the value of healthcare data: Facing an era of cyber extortion and organised crime. In *Blockchain and Clinical Trial: Securing Patient Data* 115–137 (Springer, 2019).
9. Bhosale, K.S., Nenova, M., Iliev, G.: A study of cyber attacks: In the healthcare sector. In: 2021 Sixth Junior Conference on Lighting (lighting), pp. 1–6 (2021). IEEE
10. Adat, V. & Gupta, B. B. Security in internet of things: Issues, challenges, taxonomy, and architecture. *Telecommun. Syst.* **67**, 423–441 (2018).
11. Abdullahi, M. et al. Detecting cybersecurity attacks in internet of things using artificial intelligence methods: A systematic literature review. *Electronics* **11**(2), 198 (2022).
12. Saheed, Y. K. & Arowolo, M. O. Efficient cyber attack detection on the internet of medical things-smart environment based on deep recurrent neural network and machine learning algorithms. *IEEE Access* **9**, 161546–161554 (2021).
13. Sah, A., Bhushan, B. & Shetty, C. A comprehensive study on artificial intelligence (ai) driven internet of healthcare things (ioht). In: *International conference on intelligent systems in computing and communication* 254–269 (Springer, 2023).
14. Algethami, S. A. & Alshamrani, S. S. A deep learning-based framework for strengthening cybersecurity in internet of health things (ioht) environments. *Appl. Sci.* **14**(11), 4729 (2024).
15. Kumar, M., Kim, C., Son, Y., Singh, S. K. & Kim, S. Empowering cyberattack identification in ioht networks with neighborhood-component-based improvised long short-term memory. *IEEE Internet Things J.* **11**(9), 16638–16646. <https://doi.org/10.1109/JIOT.2024.3354988> (2024).
16. Al-Azzani, A. M., Abdullah Al-Shalabi, A., Ghurab, M., Alhrazi, F., Alhomdy, S. A. & Algabri, M. Image-based feature representation for enhanced cyber-attack detection in ioht environments. In: *2024 1st international conference on emerging technologies for dependable internet of things (ICETI)* 1–11 (2024). <https://doi.org/10.1109/ICETI63946.2024.10777210>

17. Al Abdulwahid, A. Detection of middlebox-based attacks in healthcare internet of things using multiple machine learning models. *Comput. Intell. Neurosci.* **2022**(1), 2037954 (2022).
18. Vijayakumar, K. P., Pradeep, K., Balasundaram, A. & Prusty, M. R. Enhanced cyber attack detection process for internet of health things (ioht) devices using deep neural network. *Processes* **11**(4), 1072 (2023).
19. Kumar, M. & Kim, S. Securing the internet of health things: Embedded federated learning-driven long short-term memory for cyberattack detection. *Electronics* **13**(17), 3461 (2024).
20. Algethami, S. A. & Alshamrani, S. S. A deep learning-based framework for strengthening cybersecurity in internet of health things (ioht) environments. *Appl. Sci.* **14**(11), 4729 (2024).
21. Kumar, M., Singh, S. K. & Kim, S. Hybrid deep learning-based cyberthreat detection and iomt data authentication model in smart healthcare. *Futur. Gener. Comput. Syst.* **166**, 107711 (2025).
22. Karthickkumar, A., Rathnamala, S. & Prabhnanthakumar, M. Design and development of a novel cnn-bilstm hybrid deep learning model for cyber attacks detection in an ioht environment. In: *2025 IEEE international conference on interdisciplinary approaches in technology and management for social innovation (IATMSI)*, vol. 3, 1–6 (2025). <https://doi.org/10.1109/IATMSI64286.2025.10985456>
23. Rodge, S., Potey, S., Farooqui, S. & Dasarwar, P. Identification of cyberattacks in iot-powered healthcare. In: *2024 international conference on information science and communications technologies (ICISCT)* 247–252 (2024). <https://doi.org/10.1109/ICISCT64202.2024.10956726>
24. Liyakathunisa, Syeda, Z. R. & Syed, R. S. Cyber attack detection for internet of health things through federated deep learning technique. In: *2024 annual computer security applications conference workshops (ACSAC Workshops)* 194–200 (2024). <https://doi.org/10.1109/ACSACW65225.2024.00028>
25. Ji, R., Kumar, N. & Padha, D. Optimized intrusion detection approach for cyber-physical system using meta-learning with stacked generalization: An ensemble learning inspired approach. *Secur. Priv.* **8**(3), 70031 (2025).
26. Ji, R., Kumar, N. & Padha, D. Cnn-gwo-voting & hybrid: Ensemble learning inspired intrusion detection approaches for cyber-physical systems. *Proc. Indian Natl. Sci. Acad.* **91**, 1–15 (2024).
27. Shaikh, J. A. et al. Memory feedback transformer based intrusion detection system for iomt healthcare networks. *Internet Things* **32**, 101597 (2025).
28. Shaikh, J. A., Wang, C., Owais, M., Zia, U., Sima, M. W. U. & Arshad, M. Foid: A feature-optimized intrusion detection system for securing iomt healthcare networks. In: *2024 18th international conference on open source systems and technologies (ICOSST)* 1–7 (2024). <https://doi.org/10.1109/ICOSST64562.2024.10871156>
29. Faruqi, N. et al. SafetyMed: A novel iomt intrusion detection system using CNN-LSTM hybridization. *Electronics* **12**(17), 3541 (2023).
30. Al-Abadi, A. A. J., Mohamed, M. B. & Fakhfakh, A. Enhanced random forest classifier with k-means clustering (erf-kmc) for detecting and preventing distributed-denial-of-service and man-in-the-middle attacks in internet-of-medical-things networks. *Computers* **12**(12), 262 (2023).
31. Ahmed, M., Byreddy, S., Nutakki, A., Sikos, L. F. & Haskell-Dowland, P. Ecu-ioht: A dataset for analyzing cyberattacks in internet of health things. *Ad Hoc Netw.* **122**, 102621 (2021).
32. Hady, A. A., Ghubaish, A., Salman, T., Unal, D. & Jain, R. Intrusion detection system for healthcare systems using medical and network data: A comparison study. *IEEE Access* **8**, 106576–106584 (2020).
33. Mosaiyebzadeh, F. et al. Privacy-preserving federated learning-based intrusion detection system for ioht devices. *Electronics* **14**(1), 67 (2024).
34. Mosaiyebzadeh, F., Pouriyeh, S., Parizi, R. M., Han, M. & Batista, D. M. Intrusion detection system for ioht devices using federated learning. In: *IEEE INFOCOM 2023 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)* 1–6 (2023). <https://doi.org/10.1109/INFOCOMWKSHPS57453.2023.10225932>

Acknowledgements

This research is supported by Cybersecurity Research and Innovation Pioneers Grants Initiative RDIA Riyadh Saudi Arabia under Grant Number [CRPG-25-3183]. This research was also supported by Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R235). The authors would also like to acknowledge the support of AIDA Lab, Prince Sultan University, Riyadh Saudi Arabia.

Author contributions

T.S. performed data curation, formal analysis, designed methodology and wrote the original draft. A.R. dealt with software, supervision and performed visualization. M.M. conceived the idea, performed data curation, visualization, wrote the original draft, review and edit the manuscript and formal analysis. A.A.R. carried out project administration, validation, review and edit the manuscript and formal analysis. S.A.O. performed visualization and initial investigation. A.Y. performed visualization and initial investigation, supervised the study and review and edit the manuscript. All authors read and approved the final manuscript.

Funding

This research is supported by Cybersecurity Research and Innovation Pioneers Grants Initiative RDIA Riyadh Saudi Arabia under Grant Number [CRPG-25-3183]. This research was also supported by Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R235).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to A.Y.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025